

Lab 00 - Introduction to Linux: Basics of Working in the Command Line

Objective

The main objective to learn the basics of Linux command line environment. This exercise includes user creation, configuration of the system, file directory operations and permission and SSH related preparations.

Initial Steps

Open the kali linux terminal and open the default user to lab su-tpme and added newpassword for the new user and confirmed the directory with pwd and listed current content direct using ls and ls -l command .I have used parent directory using 'cd..'.Displayed the available user inside/home.

Result

A new user tpme was created and added to the sudo group to allow administrative access. The system was then updated using apt update.

```
(kali@kali)-[~/cybersecurity/CS-Vaje/Lab00]
$ whoami
kali
```

Environment Set up

The tmate package was installed, and the SSH service was started. Basic Linux commands were also tested to verify that the system was working properly.

Overall, the Kali Linux environment was successfully configured and ready for the lab activities.

```
(kali@kali)-[~/Lab00]
$ sudo adduser tpme
[sudo] password for kali:
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for tpme
Enter the new value, or press ENTER for the default
  (blank) Full Name []: tpme
  (blank) Room Number []: tpme
  (blank) Work Phone []: tpme
  (blank) Home Phone []: 233
  (blank) Other []: 678
Is the information correct? [Y/n] y
```

Basic Command-Line Tasks

1. Navigation the System

I used basic Linux navigation commands to explore and move through the filesystem. The pwd command was used to display my current working directory, while ls and ls -l were used to view the files and folders, including detailed information about them. The cd .. command allowed me to move from the current directory to its parent directory.

```
(kali@kali)-[~/lab00]
$ sudo usermod -aG sudo tpme 12, 'cracked' - home)
$ sudo visudo
visudo: /etc/sudoers.tmp unchanged
$ sudo service ssh start
$ sudo apt update
Get:1 http://kali.mirror.garr.it/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.mirror.garr.it/kali kali-rolling/main amd64 Packages [21.4 MB]
Get:3 http://kali.mirror.garr.it/kali kali-rolling/main amd64 Contents (deb) [53.2 MB]
Get:4 http://kali.mirror.garr.it/kali kali-rolling/contrib amd64 Packages [114 kB]
Get:5 http://kali.mirror.garr.it/kali kali-rolling/contrib amd64 Contents (deb) [265 kB]
Get:6 http://kali.mirror.garr.it/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:7 http://kali.mirror.garr.it/kali kali-rolling/non-free amd64 Contents (deb) [915 kB]
Fetched 76.1 MB in 30s (2,545 kB/s)
838 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Result

I successfully checked my current location, viewed the contents of directories, and navigated to the parent directory.

2. Working with Files and Directories

I practiced creating and managing files and directories in Linux. First, I created a directory named linux-exercise and moved into it. Then, I created a text file called description.txt and edited its contents using the nano text editor. After that, I created another directory named test and moved the text file into it.

Result

I successfully created directories and a text file, edited the file, and moved it to another folder.

```
$ sudo apt install tmate
The following packages were automatically installed and are no longer required:
aardvark-dns libblake3-0 libslirp0 node-readaddrp
buildah libcriu2 libsodium23 node-set-immediate-shim
catatonit libgo-14-dev libsubid5 openjdk-21-jre
common libgo23 libteamctl0 passt
containers-storage libio-pty-perl libunibreak6 podman
crun libipc-run-perl libxmlsec1-1 postgresql-common-dev
docker-compose libiptcdata0 libxmlsec1-openssl python3-pluginbase
fuse-overlayfs libmbedcrypto16 linux-base-6.18.12+kali-amd64 python3-pysnmp4
gccgo-14 libpostal-data linux-image-6.18.12+kali-amd64 python3-tz
gccgo-14-x86-64-linux-gnu libpostal1 netavark slirp4netns
golang-github-containers-common libraw23t64 node-async-each uidmap
golang-github-containers-image libstdl2-classic node-is-binary-path
libavc1394-0 libsimdutf31 node-path-is-absolute
Use 'sudo apt autoremove' to remove them.

Installing:
tmate

Installing dependencies:
libmsgpack-c2

Summary:
Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 73
Download size: 307 kB
Space needed: 803 kB / 406 GB available
```

Basic Regular commands

I practiced creating and managing files and directories in Linux. I created a directory named linux-exercise using mkdir and moved into it with cd. Then, I created a text file named description.txt using touch and edited it with the nano editor. After that, I created another directory called test and moved the text file into it using the mv command. This exercise helped me understand basic file and directory management in Linux.

```
(kali㉿kali)-[~/lab00]
└─$ su - tpme
Password:
(tpme㉿kali)-[~]
└─$ pwd
/home/tpme
└─$ ls
analysier.py
(tpme㉿kali)-[~]
└─$ ls -l
total 0
-rw-r--r-- 1 tpme tpme 0 Nov 13 13:51 analyser.py
(tpme㉿kali)-[~]
└─$ cd ..
(tpme㉿kali)-[/home]
└─$ ls
kali  tpme
(tpme㉿kali)-[/home]
└─$
```

```
(tpme㉿kali)-[/home]
└─$ sudo mkdir linux-exercise
[sudo] password for tpme:
Sorry, try again.
[sudo] password for tpme:
(tpme㉿kali)-[/home]
└─$ cd linux-exercise
(tpme㉿kali)-[/home/linux-exercise]
└─$ sudo touch descriptio.txt
(tpme㉿kali)-[/home/linux-exercise]
└─$ sudo nano descriptio.txt
descriptio.txt
(tpme㉿kali)-[/home/linux-exercise]
└─$ sudo mkdir test
(tpme㉿kali)-[/home/linux-exercise]
└─$ ls
descriptio.txt  descriptio.txt  test
(tpme㉿kali)-[/home/linux-exercise]
└─$ sudo mv descriptio.txt test/
(tpme㉿kali)-[/home/linux-exercise]
└─$ ls
descriptio.txt  test
(tpme㉿kali)-[/home/linux-exercise]
└─$ cd test/
(tpme㉿kali)-[/home/linux-exercise/test]
└─$ ls
descriptio.txt
(tpme㉿kali)-[/home/linux-exercise/test]
└─$ cat descriptio.txt
(tpme㉿kali)-[/home/linux-exercise/test]
└─$ ls
```